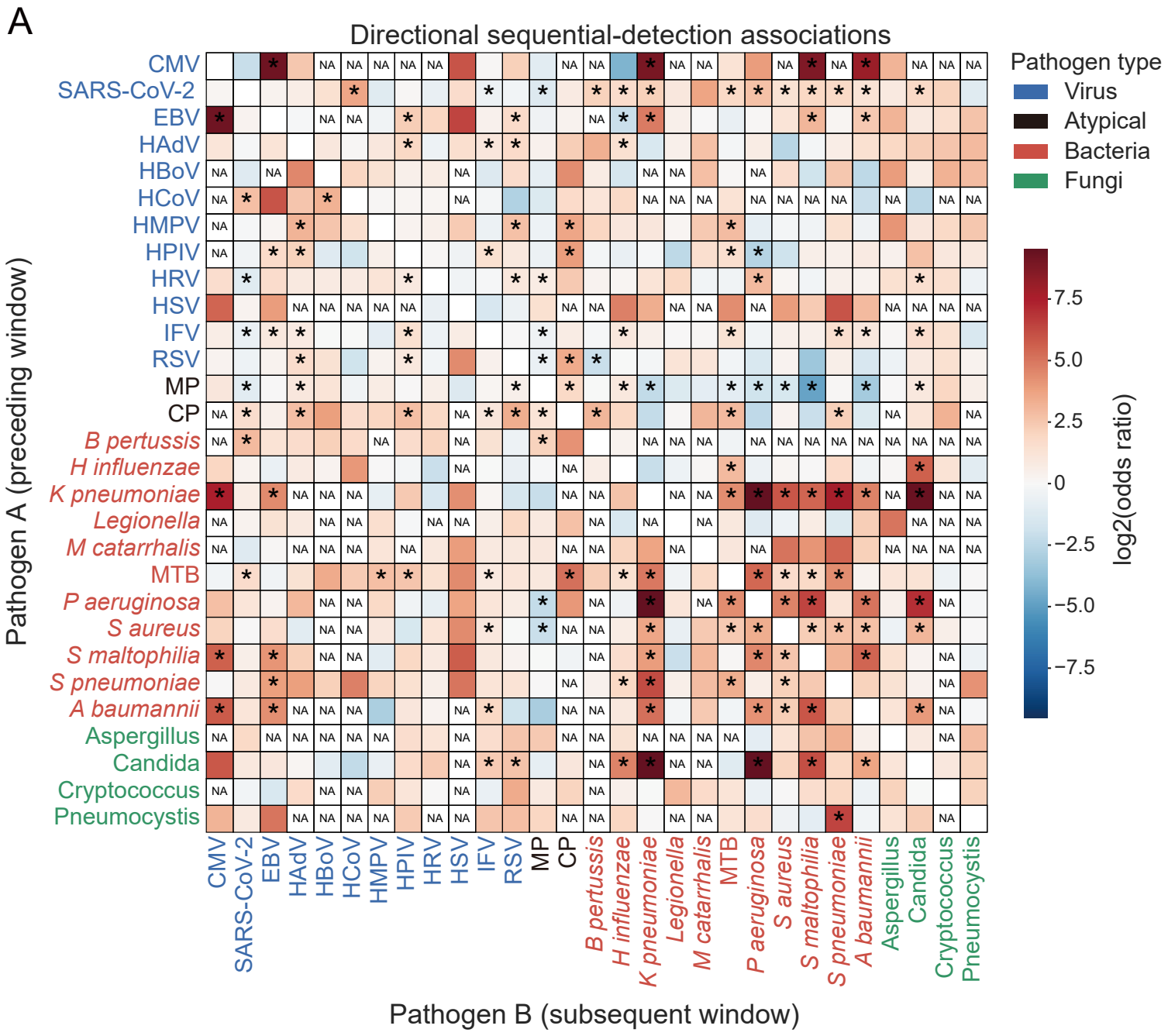

A
Directional sequential-detection associations
Pathogen type
Virus
Atypical
Bacteria
Fungi
log2(odds ratio)
7.5
5.0
2.5
0
−2.5
−5.0
−7.5
Pathogen A (preceding window)
Pathogen B (subsequent window)
CMV
SARS-CoV-2
EBV
HAdV
HBoV
HCoV
HMPV
HPIV
HRV
HSV
IFV
RSV
MP
CP
B pertussis
H influenzae
K pneumoniae
Legionella
M catarrhalis
MTB
P aeruginosa
S aureus
S maltophilia
S pneumoniae
A baumannii
Aspergillus
Candida
Cryptococcus
Pneumocystis

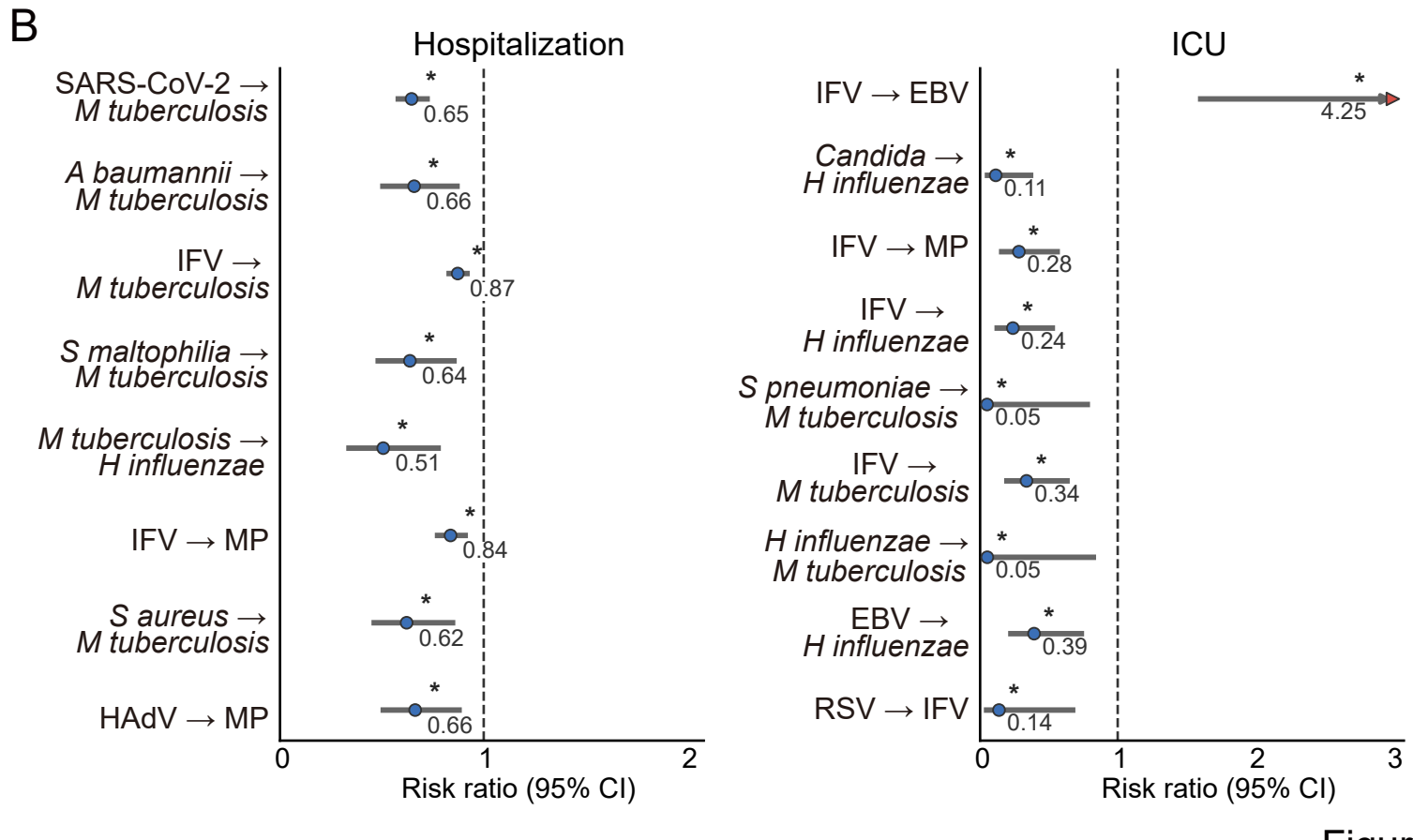

B
Hospitalization
SARS-CoV-2 → M tuberculosis
0.65
A baumannii → M tuberculosis
0.66
IFV → M tuberculosis
0.87
S maltophilia → M tuberculosis
0.64
M tuberculosis → H influenzae
0.51
IFV → MP
0.84
S aureus → M tuberculosis
0.62
HAdV → MP
0.66
0
1
2
Risk ratio (95% CI)
ICU
IFV → EBV
4.25
Candida → H influenzae
0.11
IFV → MP
0.28
IFV → H influenzae
0.24
S pneumoniae → M tuberculosis
0.05
IFV → M tuberculosis
0.34
H influenzae → M tuberculosis
0.05
EBV → H influenzae
0.39
RSV → IFV
0.14
0
1
2
3
Risk ratio (95% CI)

Figure 6